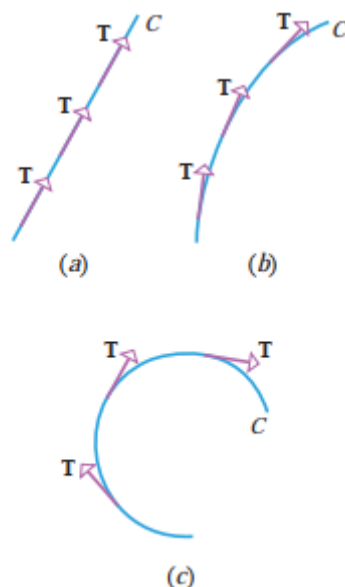


Curvature and Radius of Curvature of a Curve



▲ Figure 12.5.1

Suppose that C is the graph of a smooth vector-valued function in 2-space or 3-space that is parametrized in terms of arc length. Figure 12.5.1 suggests that for a curve in 2-space the “sharpness” of the bend in C is closely related to $d\mathbf{T}/ds$, which is the rate of change of the unit tangent vector $\mathbf{T}$ with respect to s . (Keep in mind that $\mathbf{T}$ has constant length, so only its direction changes.) If C is a straight line (no bend), then the direction of $\mathbf{T}$ remains constant (Figure 12.5.1a); if C bends slightly, then $\mathbf{T}$ undergoes a gradual change of direction (Figure 12.5.1b); and if C bends sharply, then $\mathbf{T}$ undergoes a rapid change of direction (Figure 12.5.1c).

Formulas for Curvature

- Cartesian:

$$\kappa = \frac{|y''|}{(1 + (y')^2)^{3/2}}$$

- Parametric:

$$\kappa = \frac{|x'y'' - y'x''|}{((x')^2 + (y')^2)^{3/2}}$$

- Polar:

$$\kappa = \frac{|r^2 + 2(r')^2 - rr''|}{(r^2 + (r')^2)^{3/2}}$$

Find the curvature of $y = x^2$ at $x = 1$.

$$y' = 2x, \quad y'' = 2$$

At $x = 1$:

$$\kappa = \frac{|2|}{(1 + (2)^2)^{3/2}} = \frac{2}{(5)^{3/2}} = \frac{2}{5\sqrt{5}}$$

Find curvature of the parabola in parametric form

$$x = t, \quad y = t^2$$

(at $t = 1$)

$$x' = 1, \quad x'' = 0$$

$$y' = 2t, \quad y'' = 2$$

At $t = 1$:

$$\kappa = \frac{|1 \cdot 2 - (2)(0)|}{(1^2 + 2^2)^{3/2}} = \frac{2}{5^{3/2}}$$

Find the curvature of the cardioid

$$r = a(1 + \cos \theta)$$

at $\theta = 0$.

Step 1: Compute derivatives

$$r = a(1 + \cos \theta)$$

$$r' = -a \sin \theta$$

$$r'' = -a \cos \theta$$

Step 2: Evaluate at $\theta = 0$

$$r(0) = a(1 + 1) = 2a$$

$$r'(0) = 0$$

$$r''(0) = -a$$

Step 3: Substitute into curvature formula

Numerator:

$$|r^2 + 2(r')^2 - rr''| = |(2a)^2 + 0 - (2a)(-a)| = |4a^2 + 2a^2| = 6a^2$$

Denominator:

$$(r^2 + (r')^2)^{3/2} = (4a^2)^{3/2} = 8a^3$$

Final Answer

$$\kappa(0) = \frac{6a^2}{8a^3} = \frac{3}{4a}$$

Practice Questions

1. Find the curvature of $y = x^3$ at $x = 1$.
2. Find the curvature of $y = \ln x$ at $x = 1$.
3. Find the curvature of $y = \sin x$ at $x = 0$.

4. For $x = t^2$, $y = t^3$, find curvature at $t = 1$.

5. Find curvature of the circle

$$x = a \cos t, \quad y = a \sin t$$

at any t .

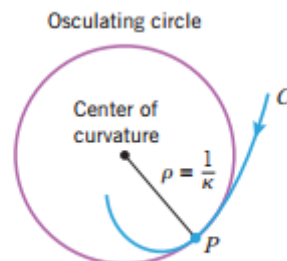
6. For $x = t$, $y = \sin t$, find curvature at $t = 0$.

7. Find the curvature of the cardioid $r = a(1 + \cos \theta)$ at $\theta = 0$.

8. Find the curvature of the spiral $r = a\theta$ at $\theta = \pi$.

9. Show that for $r = a$, curvature is constant.

Radius of Curvature



▲ Figure 12.5.6

In general, if a curve C in 2-space has nonzero curvature κ at a point P , then the circle of radius $\rho = 1/\kappa$ sharing a common tangent with C at P , and centered on the concave side of the curve at P , is called the *osculating circle* or *circle of curvature* at P (Figure 12.5.6). The osculating circle and the curve C not only touch at P but they have equal curvatures at that point. In this sense, the osculating circle is the circle that best approximates the curve C near P . The radius ρ of the osculating circle at P is called the *radius of curvature* at P , and the center of the circle is called the *center of curvature* at P (Figure 12.5.6).